phases of water can coexist in equilibrium at 273.16 K. *Kelvin is defined as 1/273.16 of the triple point temperature.* The formula for converting a Celsius scale to Kelvin scale is: K = C + 273.15

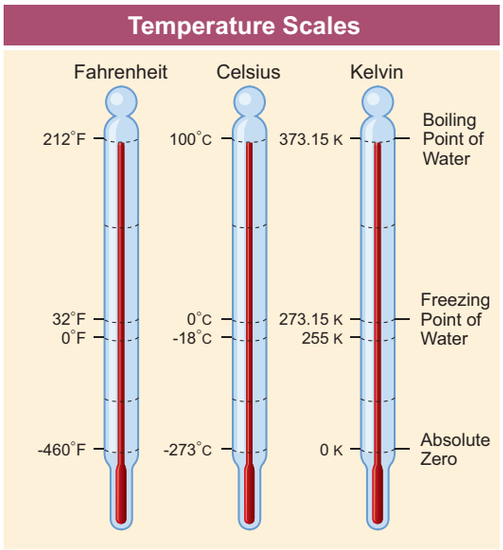


Figure 7.6 Types of temperature scales

Absolute zero

The temperature at which the pressure and volume of a gas theoretically reaches zero is called absolute zero. This is shown in Figure 7.7.

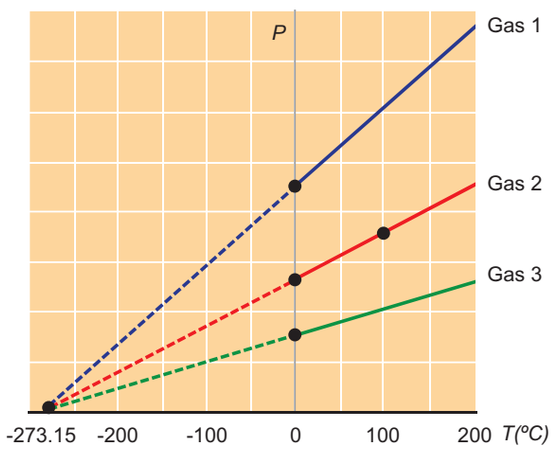


Figure 7.7 Variation of pressure (P) with temperature (T).

For all gases, the pressure extrapolates to zero at the temperature –273.15 °C. It is known as absolute zero or 0 K. Some base line temperatures in the three temperature scales are shown in Table 7.1.

Table. 7.1 Baseline temperatures in three scales.

Temperature	Kelvins (K)	Degrees Celcius (°C)	Degrees Fahrenheit (°F)
Boiling point of water	373.15	100	212
Melting point of ice	273.15	0	32
Absolute zero	0	–273	–460

Problem 1

Convert the following

i. 25 °C to Kelvin ii. 200 K to °C

Solution:

i. $T_K = T_{°C} + 273.15$

$T_K = 25 + 273.15 = 298.15$ K

ii. $T_{°C} = T_K - 273.15$

$T_{°C} = 200 - 273.15 = -73.15$ °C

Problem 2

Convert the following

i. 35° C to Fahrenheit (°F) ii. 14 °F to °C

Solution:

i. $T_{°F} = T_{°C} \times 1.8 + 32$

$T_{°F} = 25° \text{ C} \times 1.8 + 32 = 77$ °F

ii. $T_{°C} = (T_{°F} - 32)/ 1.8$

$T_{°C} = (14°\text{F} - 32)/ 1.8 = -10$ °C

7.4 Specific Heat Capacity

You might have felt that the land is cool in the morning and hot during day time. But, water in a lake will be almost at a particular temperature both in the morning as well as in the afternoon. Both are subjected to same

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